

Physics of Stars III

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1 Star Formation

The gravitational potential energy (E_{GR}) is defined as

$$E_{GR} = -f \frac{GM^2}{R}$$

Where $f = \frac{3}{5}$ or $f = 1$. The internal thermal energy (E_{IN}) for N particles is

$$E_{IN} = \frac{3}{2} NkT$$

For collapse to occur, the magnitude of gravitational energy must exceed the internal energy

$$|E_{GR}| \geq E_{IN} \Rightarrow \frac{GM^2}{R} \geq \frac{3}{2} NkT$$

The total number of particles N is given by the total mass M divided by the average particle mass $\bar{m}$

$$N = \frac{M}{\bar{m}}$$

Substituting N into the equilibrium equation

$$\frac{GM^2}{R} = \frac{3}{2} \frac{M}{\bar{m}} kT$$

Solving for M

$$M = \frac{3kT}{2G\bar{m}} R$$

Given that the radius R relates to density ρ as

$$R = \left(\frac{3M}{4\pi\rho} \right)^{1/3}$$

Substituting R back into the mass equation

$$\frac{GM}{\left(\frac{3M}{4\pi\rho} \right)^{1/3}} = \frac{3}{2} \frac{1}{\bar{m}} kT \Rightarrow \frac{G^3 M^2 4\pi\rho}{3} = \left(\frac{3}{2} \right)^3 \left(\frac{kT}{\bar{m}} \right)^3$$

This leads to the Jeans Density

$$\rho_J = \frac{3}{4\pi M^2} \left(\frac{3kT}{2G\bar{m}} \right)^3$$

For a monatomic gas like atomic hydrogen, each particle contributes $\frac{3}{2}kT$. Each of the three translational degrees of freedom contributes $\frac{1}{2}kT$. In Giant Molecular Clouds (GMCs), we have H_2 , but at low temperatures, rotation and vibration degrees of freedom are not active.

Based on the derivations

$$\begin{aligned} \rho_J &\propto \frac{1}{M^2} \\ \rho_J &\propto T^3 \end{aligned}$$

Typical Values for GMCs are $T \approx 20$ K $M \approx 2 \times 10^{33}$ kg (approx. $1000 M_\odot$) $\rho \approx 10^{-22}$ g cm $^{-3}$

2 Protostar Formation

A protostar begins its contraction once it reaches the Jeans Mass (M_J). When the cloud reaches a certain density (approximately 10^{10} kg for certain scales), it fragments and contracts. This stage can occur independently for a 1 Solar mass ($M_\odot$) object or within a cloud with a diameter $d \sim 10^{15}$ cm. A typical protostar might have a mass of 2×10^{30} kg and a radius of approximately 10^{11} m. Contraction occurs when the magnitude of gravitational potential energy $|E_{GR}|$ is greater than the internal energy E_{IN} .

At this stage, while the object is a "hot star" in terms of temperature, no nuclear fusion is taking place yet. Gravitational energy (E_{GR}) does not heat the gas. But it is absorbed by dissociation (E_D) and ionization (E_I) processes.

- Molecular Dissociation $H_2 + \epsilon_D \rightarrow 2H$ requires 4.5 eV per molecule
- Atomic Ionization $H \rightarrow p^+ + e^-$ requires 13.6 eV per atom

Initially, the collapse is in "free-fall" (E_{FF}) because gravity is unopposed. The change in gravitational energy as the radius shrinks from R_1 to R_2 is expressed as

$$\Delta E_{GR} = E_{GR(final)} - E_{GR(initial)} = \frac{GM^2}{R_2} - \frac{GM^2}{R_1}$$

This early collapse stage lasts approximately 20,000 years ¹

2.1 Protostar Derivations

For a protostar of mass M composed of hydrogen, the number of particles N and the energy required for dissociation and ionization are defined as

- Total Hydrogen atoms: $N = \frac{M}{m_H}$
- Molecular Hydrogen (H_2): $\bar{m} = m_H$, $N = \left(\frac{M}{2m_H}\right)$
- Dissociation Energy (E_D): $E_{ED} = \left(\frac{M}{2m_H}\right) \epsilon_D$
- Ionization Energy (E_I): $E_{EI} = \left(\frac{M}{m_H}\right) \epsilon_I$

The total energy needed to dissociate and ionize the protostar is the sum of these two components

$$E_T \approx \left(\frac{M}{2m_H}\right) \epsilon_D + \left(\frac{M}{m_H}\right) \epsilon_I$$

The change in gravitational potential energy as the protostar contracts from R_1 to R_2 supplies this energy

$$\frac{GM^2}{R_2} - \frac{GM^2}{R_1} \approx \left(\frac{M}{2m_H}\right) \epsilon_D + \left(\frac{M}{m_H}\right) \epsilon_I$$

Since $R_1 \gg R_2$, the initial potential energy is negligible, leading to the approximation

$$|E_{GR}| = \frac{GM^2}{R_2} \approx \left(\frac{M}{2m_H}\right) \epsilon_D + \left(\frac{M}{m_H}\right) \epsilon_I$$

For a solar mass ($1 M_\odot$) protostar, this initial collapse phase lasts $\sim 20,000$ years and contracts to a radius $R_2 \approx 10^{11}$ m. Once the hydrogen is mostly ionized, the protostar becomes **opaque**. At this point, gravitational energy (E_{GR}) begins to convert into thermal energy rather than being "used up" by ionization. The collapse slows down as the star enters Hydrostatic Equilibrium. There are 2 particles for each hydrogen atom (proton and electron) after ionization. Kinetic Energy (E_{KE})

$$E_{KE} = \frac{3}{2} N k T = \frac{3}{2} \left(2 \frac{M}{m_H}\right) k T$$

¹Unlike protostars, normal stars heat up and stop collapsing due to hydrostatic equilibrium

Virial Theorem: $2E_{KE} + E_{GR} = 0$.

Considering dissociation (ϵ_D) and ionization (ϵ_I) energies is given by

$$3 \left(\frac{M}{m_H} \right) kT = \frac{1}{2} \left[\frac{M}{2m_H} \epsilon_D + \frac{M}{m_H} \epsilon_I \right]$$

Solving for $kT = \frac{1}{6} (\frac{1}{2} \epsilon_D + \epsilon_I)$ $kT = \frac{1}{12} (\epsilon_D + 2\epsilon_I)$ Substituting $\epsilon_D = 4.5$ eV and $\epsilon_I = 13.6$ eV

$$kT = \frac{1}{12} (4.5 + 2 \times 13.6) \approx 2.6 \text{ eV}$$

The temperature of the protostar at this stage is approximately 30,000 Kelvin, which is independent of the mass. Before this point, the protostar is in free-fall because gravity is unopposed. Initial Collapse lasts approximately 20,000 years. 30,000 K is too cold for nuclear fusion (which requires millions of K). The star enters a slow contraction phase governed by the Virial Theorem, where temperature (T) and density (ρ) gradually increase.

3 Stardom and Quantum Limits

The primary question is whether a slowly contracting protostar has sufficient mass to ignite nuclear fusion. If the mass is insufficient, the object becomes a brown dwarf. In these objects, pressure is determined by quantum mechanics rather than classical thermal pressure.

During the contraction, as density increases, the average distance between particles (d) decreases until it reaches the quantum limit. De Broglie Wavelength (λ)

$$\lambda = \frac{h}{p} = \frac{h}{(m_e kT)^{1/2}}$$

Classical mechanics remains valid only if the particle wave functions do not overlap, meaning $d \gg \lambda$. To define the limit, we look at the number density (n) and average particle mass ($\bar{m}$): Volume per particle is defined as $V \approx d^3 \approx \frac{1}{n}$. Number Density is $n = \frac{\rho}{\bar{m}}$. Distance (d)

$$d = \left(\frac{1}{n} \right)^{1/3} = \left(\frac{\bar{m}}{\rho} \right)^{1/3}$$

In the classical regime, the Virial Theorem governs the relationship between temperature, mass, and radius

$$kT \approx \frac{GM\bar{m}}{3R}$$

Given that density is $\rho = \frac{M}{V} = \frac{3M}{4\pi R^3}$, we can derive the scaling for the radius R and temperature T .

- $R \propto M^{1/3} \rho^{-1/3}$
- $T \propto \frac{M}{M^{1/3} \rho^{-1/3}} \rightarrow T \propto M^{2/3} \rho^{1/3}$

This relationship holds as long as the density ρ remains low enough for the classical limit to apply. As the protostar contracts, its density increases. The classical regime ends when the particle spacing reaches the quantum limit. The critical density (ρ_{crit}) where the gas becomes degenerate is reached when

$$\rho = \bar{m} \left(\frac{(m_e kT)^{3/2}}{h^3} \right)$$

As the protostar contracts, the temperature rises proportionally to $\rho^{1/3}$. Once the quantum limit is reached, the particles become degenerate and resist further compression. The transition occurs when the number density n satisfies the condition

$$\frac{\bar{m}}{\rho} \gg \frac{h^3}{(m_e kT)^{3/2}} \ll \frac{\bar{n}}{\lambda^3}$$

(Summary of Equations used in derivation: $2E_{KE} + E_{GR} = 0$, $E_{GR} = -\frac{GM^2}{R}$)

Mass (M)	Max Temp (T)	Classification	Outcome
$> 0.08M_{\odot}$	$> 10 \times 10^6$ K	True Star	Nuclear fusion ignites
$< 0.08M_{\odot}$	$< 10 \times 10^6$ K	Brown Dwarf	Degeneracy stops contraction before fusion

Property	Symbol	Value (SI)	Physical Context
Mass	$M_{\odot}$	$\approx 2 \times 10^{30}$ kg	-
Radius	$R_{\odot}$	$\approx 7 \times 10^8$ m	Distance from the core to the photosphere.
Average Density	$\langle \rho \rangle$	1.4×10^3 kg m $^{-3}$	Slightly denser than water.
Mean Particle Mass	$\bar{m}$	$0.6m_H$	$\bar{m} = \mu m_H$ accounts for a fully ionized plasma mix (H and He).
Age & Equilibrium	t_{age}	$\approx 4.5 \times 10^9$ years	Maintained by hydrostatic equilibrium $\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}$

4 Sun

The Sun has been in hydrostatic equilibrium for approximately 4.5 billion years.

The internal pressure is estimated by considering the gravitational potential energy over the volume. Average Pressure ($\langle P \rangle$)

$$\langle P \rangle = -\frac{1}{3} \frac{E_{GR}}{V} = \frac{GM_{\odot}^2}{4\pi R_{\odot}^4} \approx 10^{14} \text{ Pa}$$

This pressure is roughly a billion times greater than Earth's atmospheric pressure. Assuming the solar interior behaves as a classical ideal gas where $\langle P \rangle = nkT_I$

$$kT_I = \frac{GM_{\odot}\bar{m}}{3R_{\odot}} \approx 0.5 \text{ keV} \Rightarrow T_I \approx 6 \times 10^6 \text{ K}$$

4.1 The Random Walk

Although the internal temperature is millions of degrees, the surface temperature (T_E) is only 6000 K because the Sun is opaque. Radiation is trapped, scattered, and re-emitted by electrons and protons. For N steps of length l (mean free path), the total displacement D is:

$$D^2 = Nl^2$$

To escape, a photon must travel a net distance $D \sim R_{\odot}$

$$N = \frac{R_{\odot}^2}{l^2}$$

$$t_{RW} = N \left(\frac{l}{c} \right) = \frac{R_{\odot}^2}{cl}$$

For a mean free path $l \approx 1$ mm, the escape time is approximately 50,000 years. The relationship between the internal temperature (T_I) and the effective surface temperature (T_E) is derived from the luminosity ratio and the "bottleneck" effect of radiation.

$$\frac{L_{\odot}}{L_I} \approx \frac{t_{free}}{t_{RW}} = \frac{R_{\odot}/c}{R_{\odot}^2/cl} = \frac{l}{R_{\odot}}$$

Using the Stefan-Boltzmann Law ($L \propto T^4$)

$$\frac{T_E^4}{T_I^4} \approx \frac{l}{R_{\odot}} \Rightarrow T_E \approx \left[\frac{l}{R_{\odot}} \right]^{1/4} T_I$$

Substituting $T_I = 6 \times 10^6$ K yields the observed surface temperature $T_E \approx 6000$ K.